Platform Engineering Day EUROPE

AMSTERDAM, THE NETHERLANDS

23 MARCH 2026

#PLATENGDAY

How (not) To Stage Your (Kubernetes) Platform

Promotion failed

Your speaker

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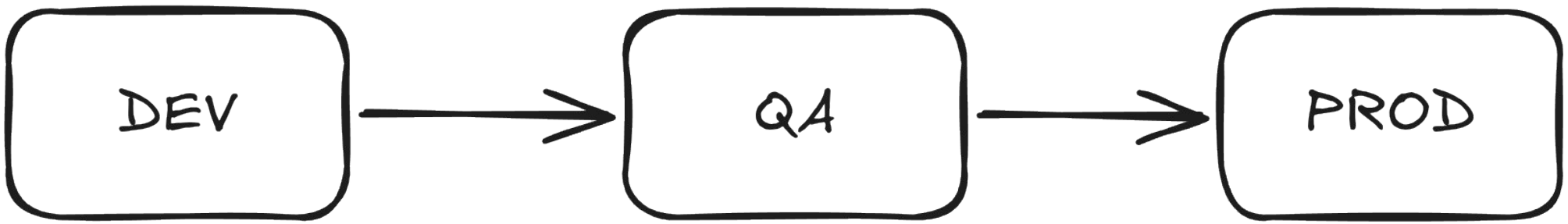
ODIT.Services

@nicolaiort

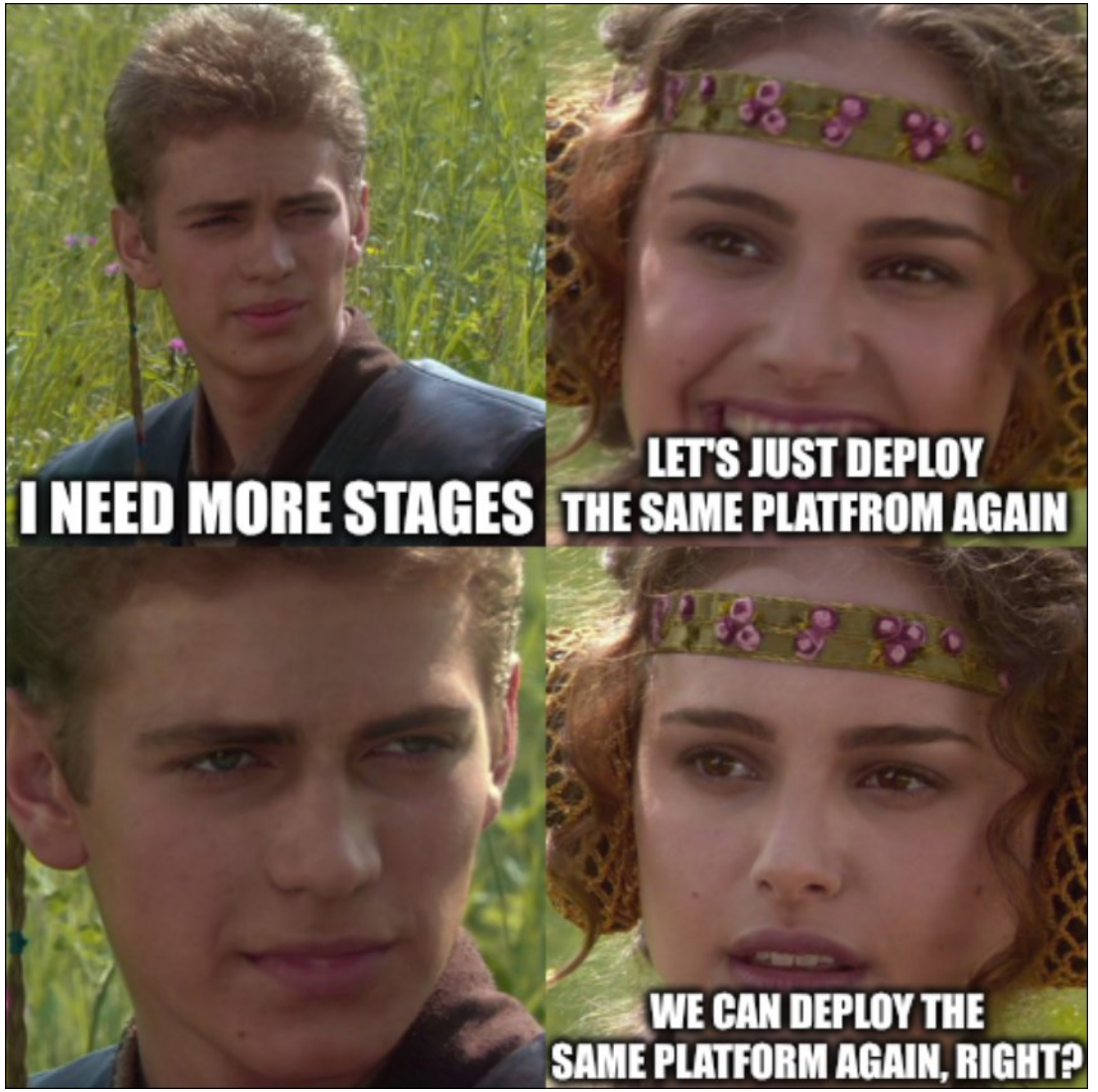
I „do cloud/platform stuff“

Staging

It‘s simple



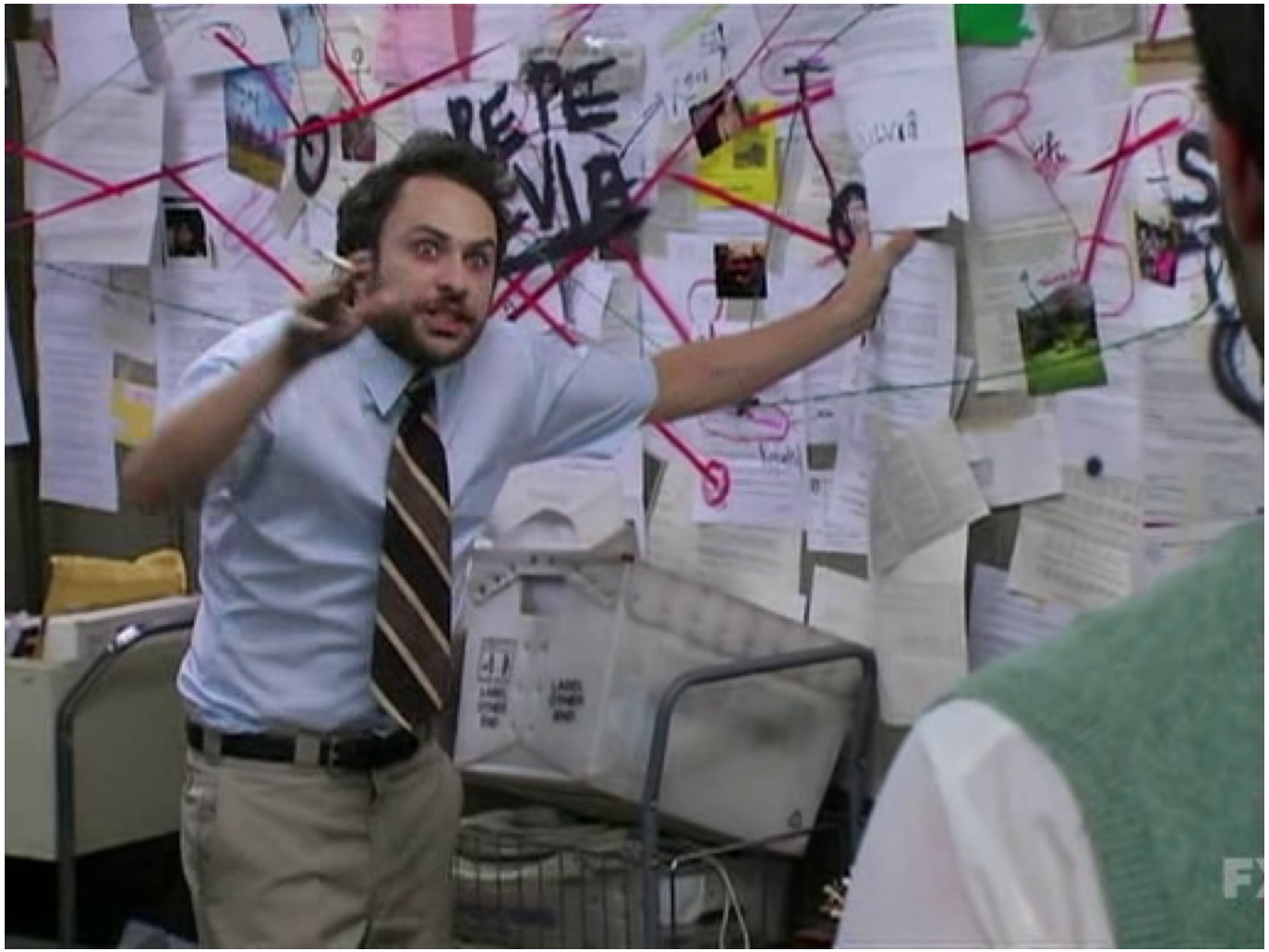
It‘s simple



isn‘t it?

What is a platform

What is a platform

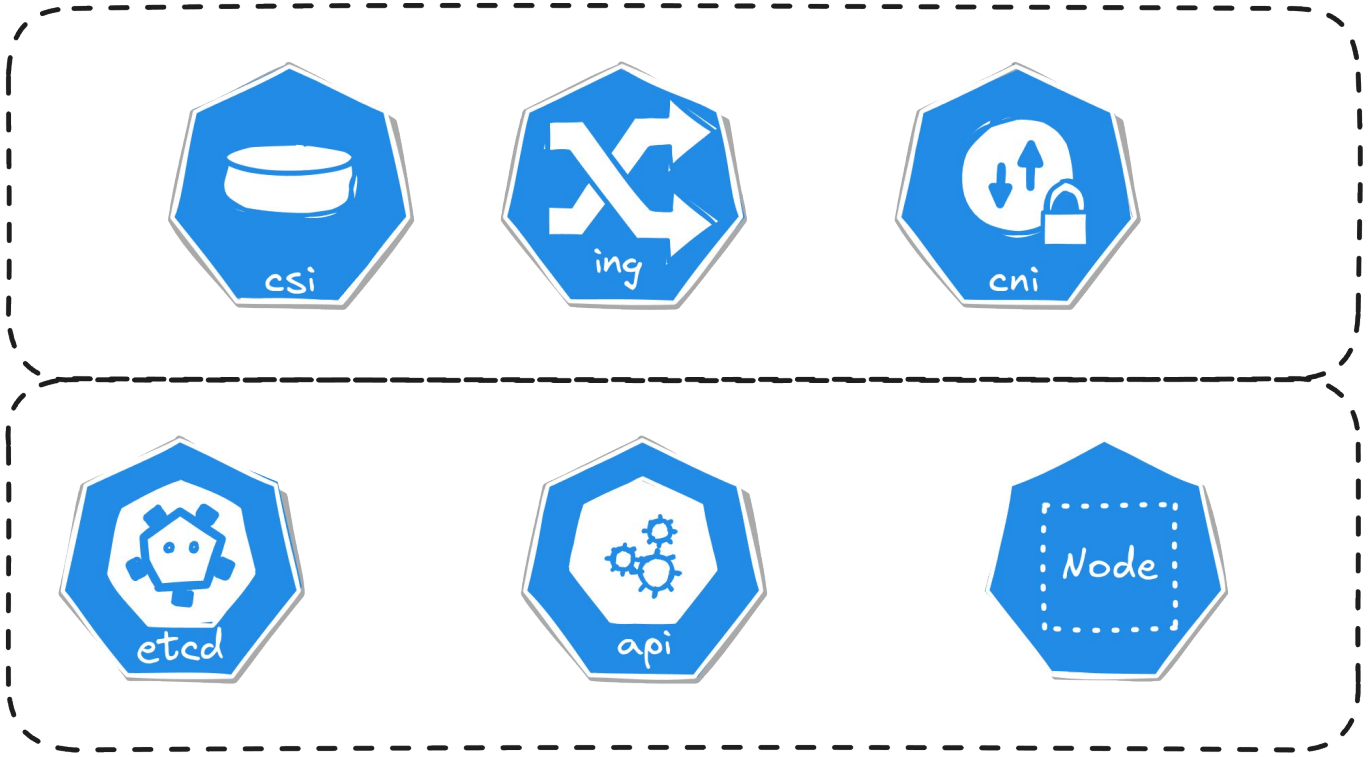


What is our platform?

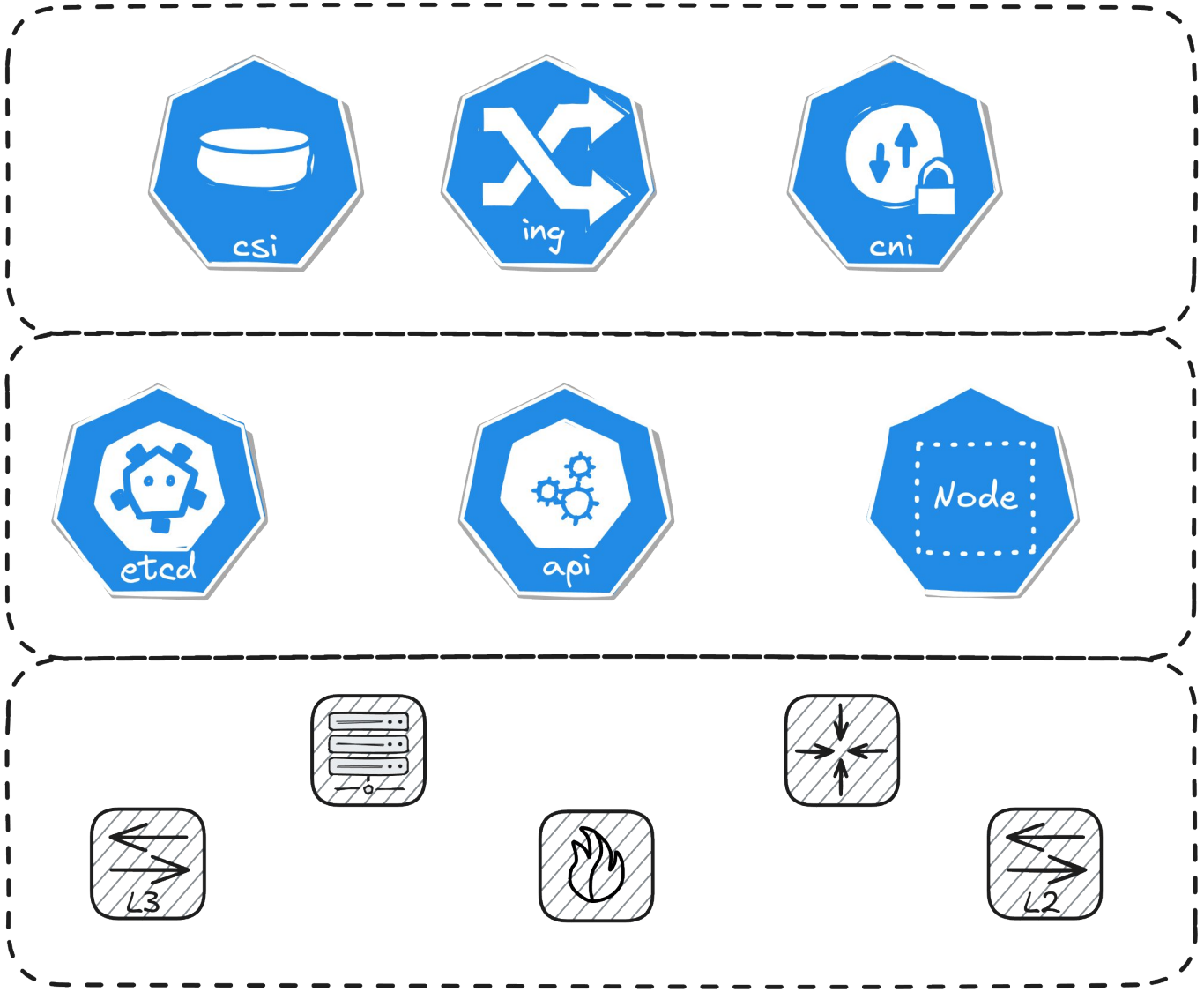
kubernetes **as a Service***

What is our platform made of

What is our platform made of



What is our platform depending on



And what about these „stages“

The misterious stage

The misterious stage

A copy of our stack

And how many do i need

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1. Developers: We need DEV to test and PROD to run our stuff

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4. Infra: Well we need to test new firmware updates

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And how many do I need

It depends!

(only cary what you can)

And how many do I need: My recommendation

1. Fullfill you customers needs
2. Add a stage for yourself
3. Try to combine stages if possible
4. Keep things reasonable


We
have a
platform


We have a platform
We have stages

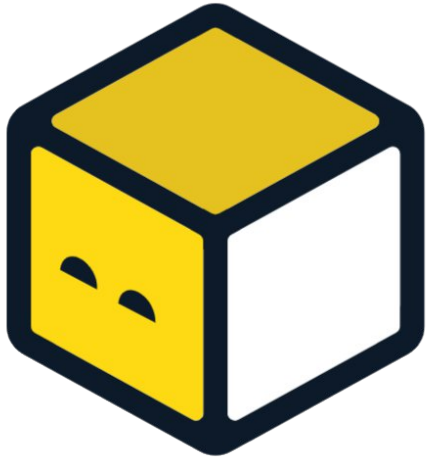

We have a platform
We have stages
We still need a state
We still need a state
imgflip.com

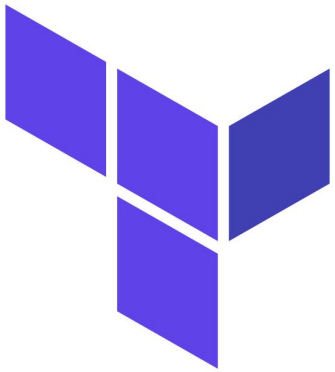
State management

Part 1: Apply state

Tooling

It does not really matter
you always need a way to apply changes





A (custom) statefile + magic

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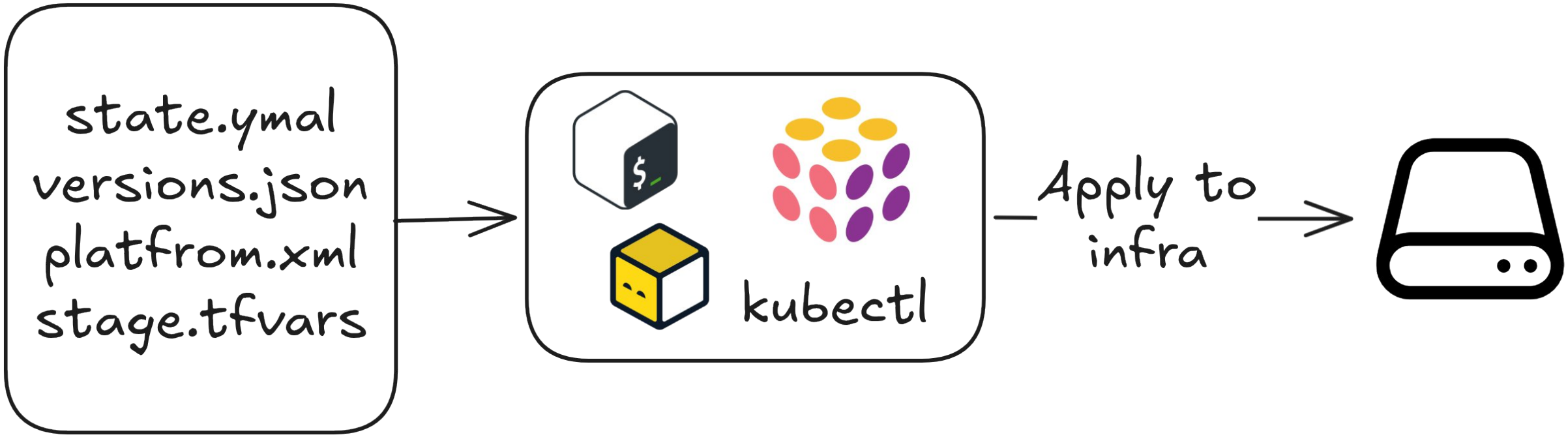
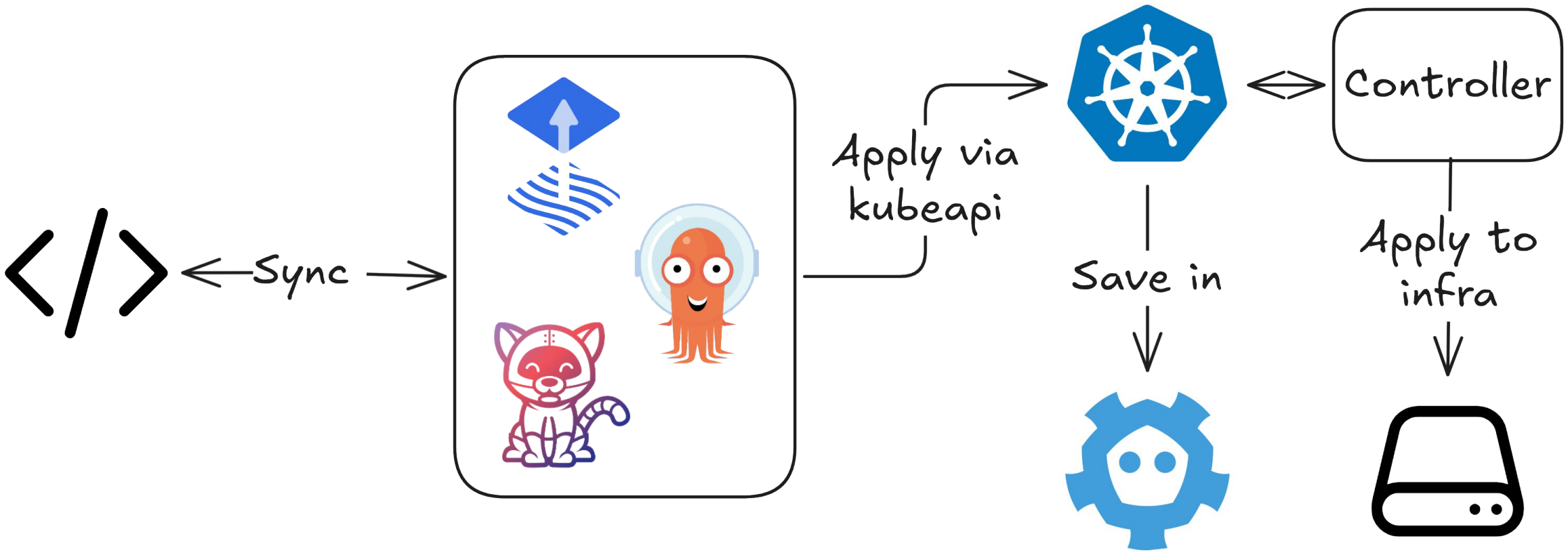


Your favorite GitOps tool + Operators

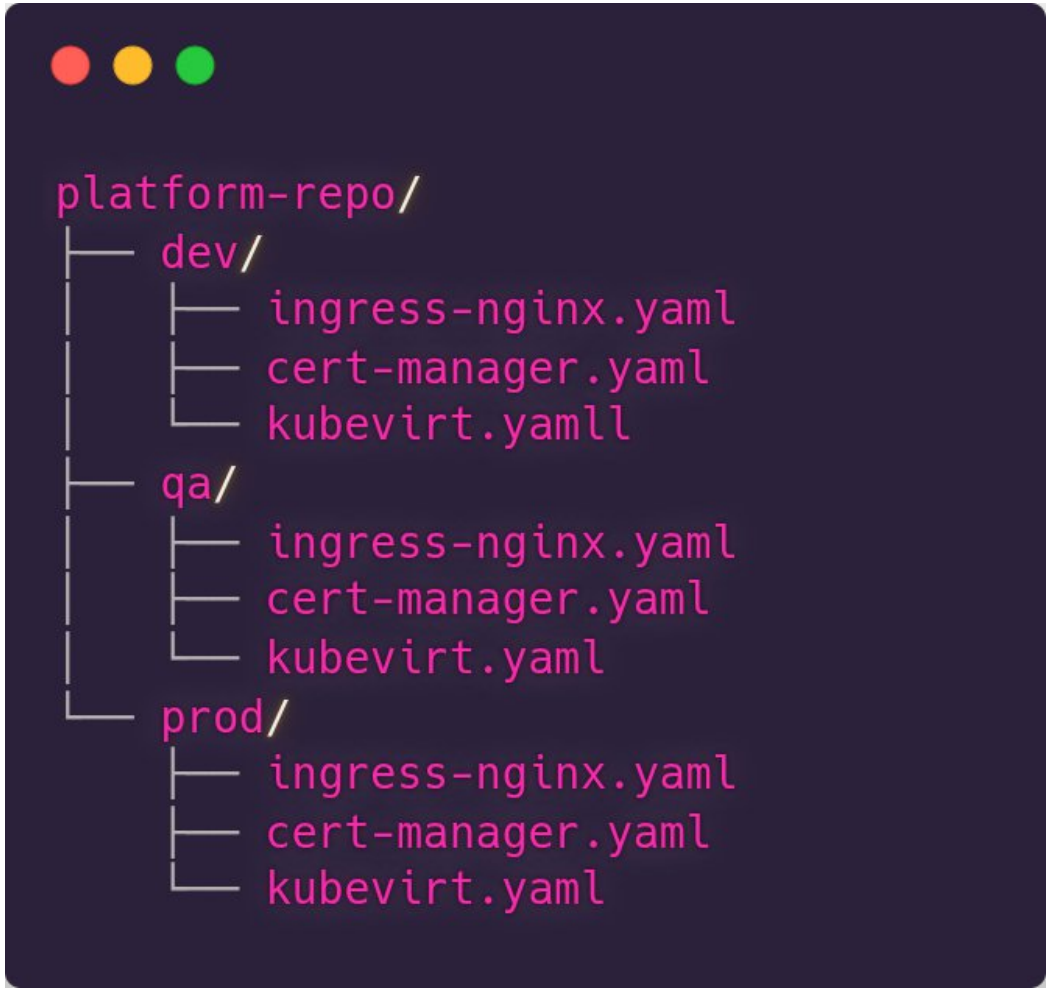


At the end of the day

State management

Part 2: Organize state

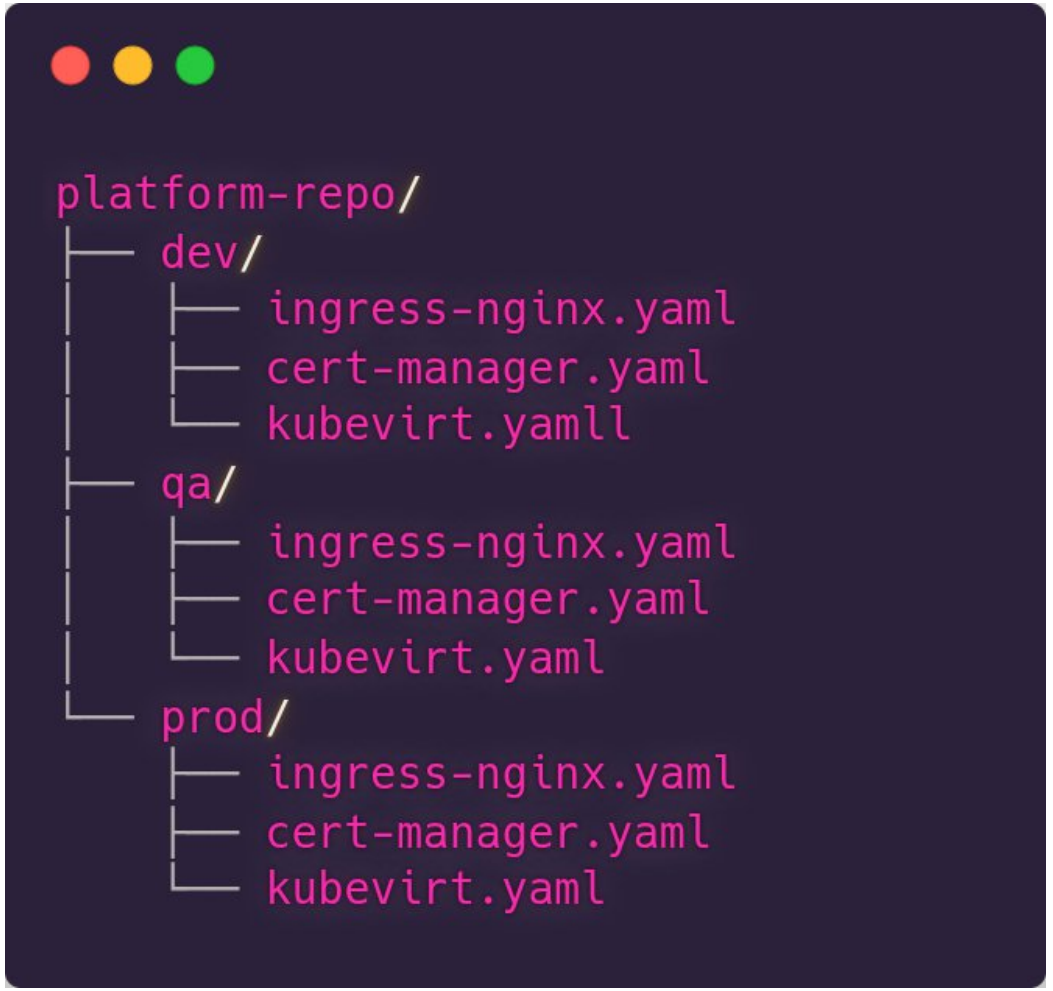
The duplication-approach



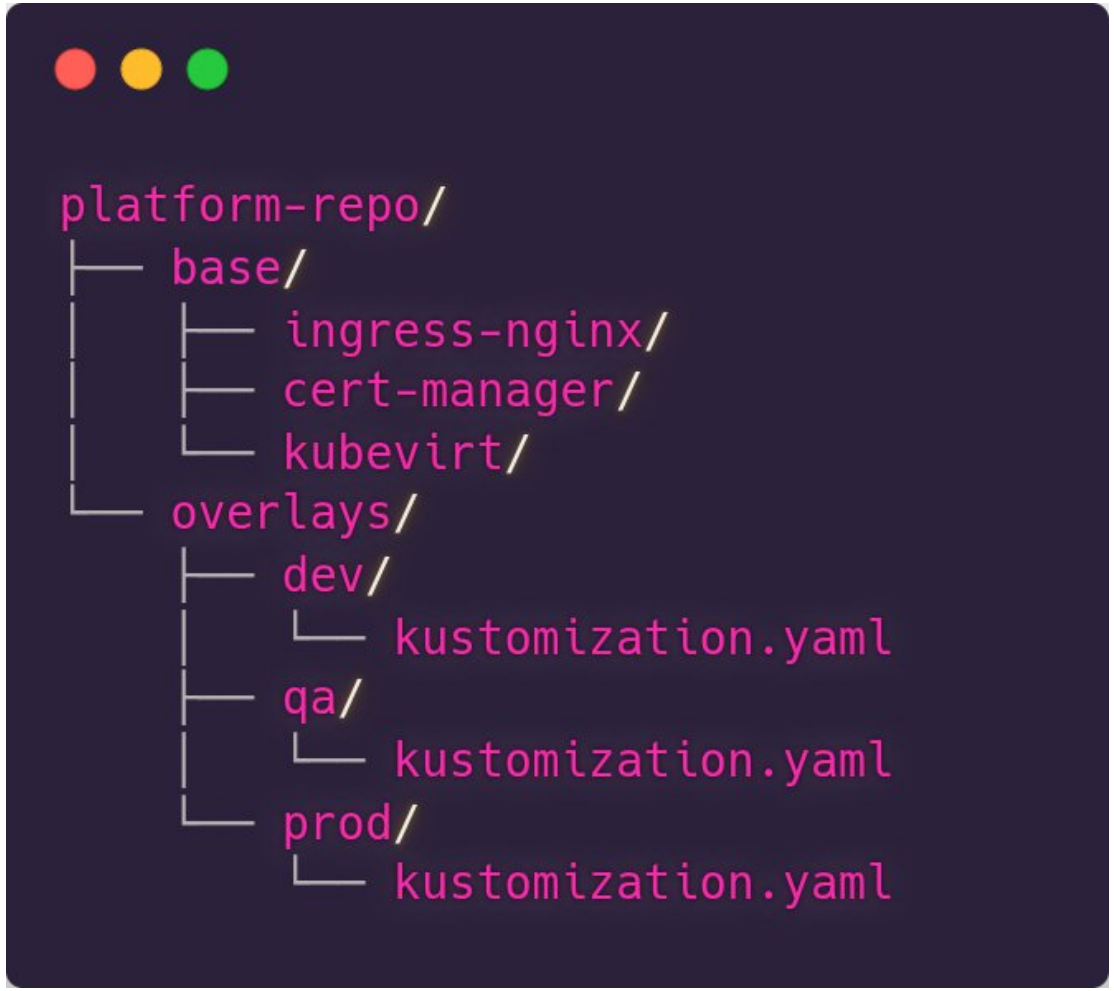
The duplication-approach

+ Small teams
+ Early start
+ Per-stage flexibility

- Duplicate files
- Hight drift risk
- Hard to scale



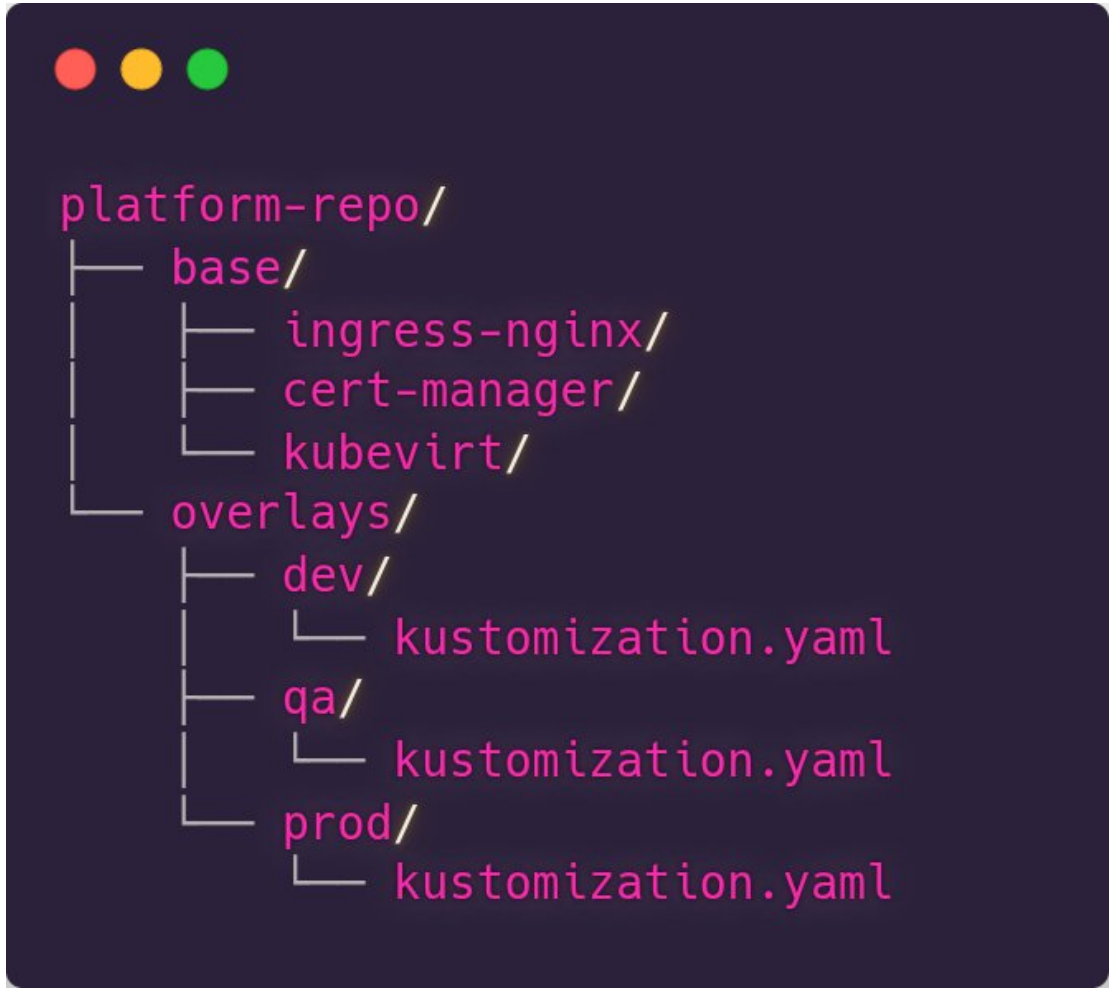
Extracting common files



Extracting common files

+ Less duplication
+ Speration of share/specific

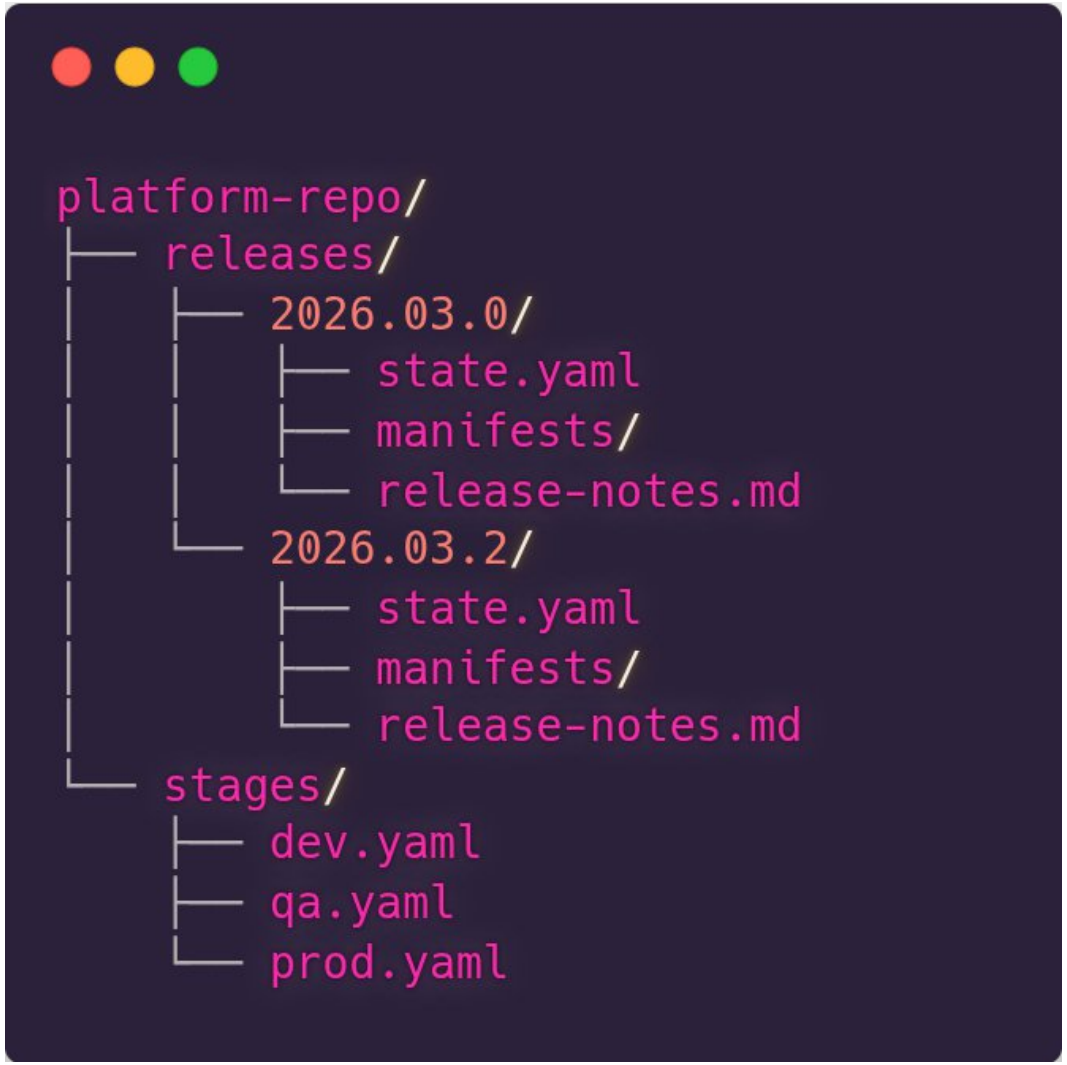
- Can become messy
- Only for simmilar stages



Git Tags

+ Clear release artifact
+ Easy to audit/diff
+ Simplifies rollback

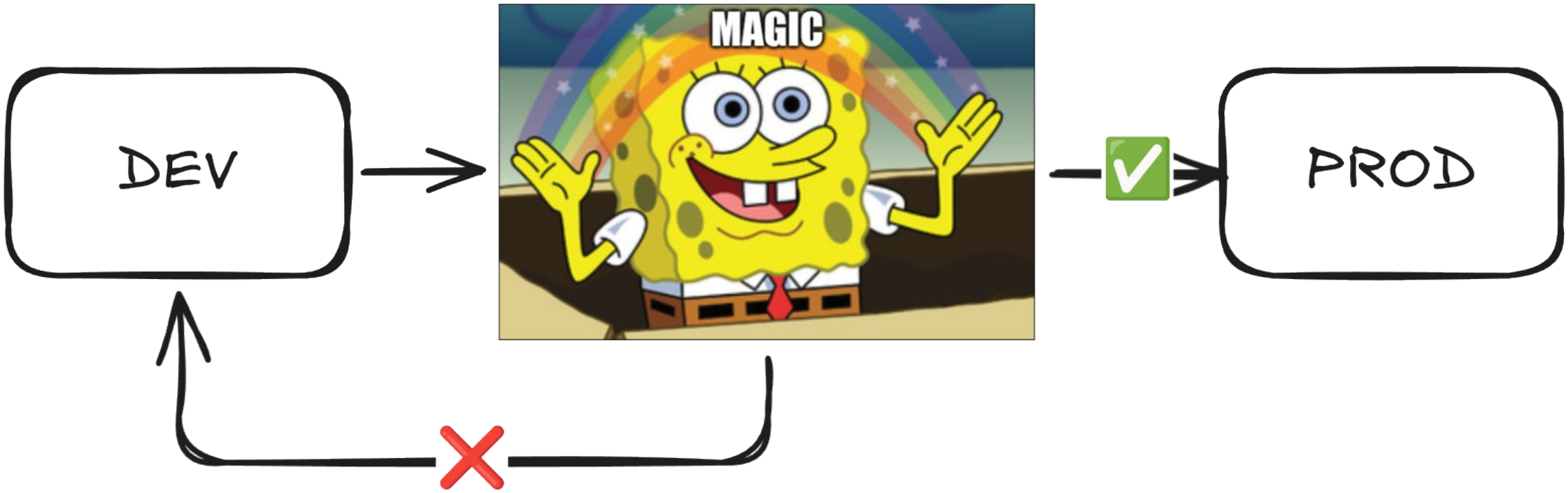
- Less flexible
- Additional engineering overhead



How do we promote state

Aka get changes from DEV to QA to PROD?

Your dreams



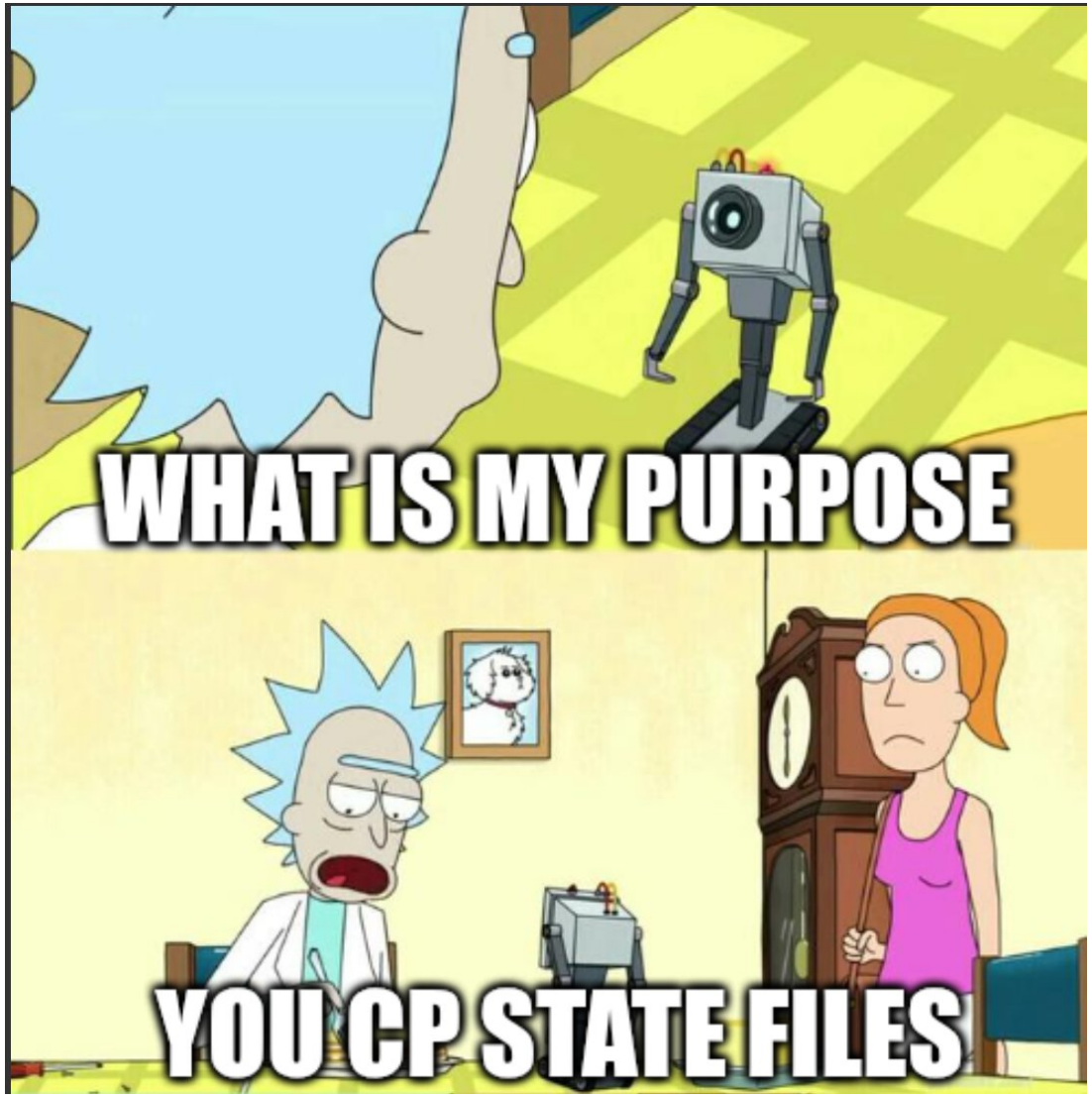
Manual promotion

Manual promotion

+ Small teams
+ Early days

- Compliance
- Scale
- Consistency

Bare minimum: automatic promotion



Static Delays

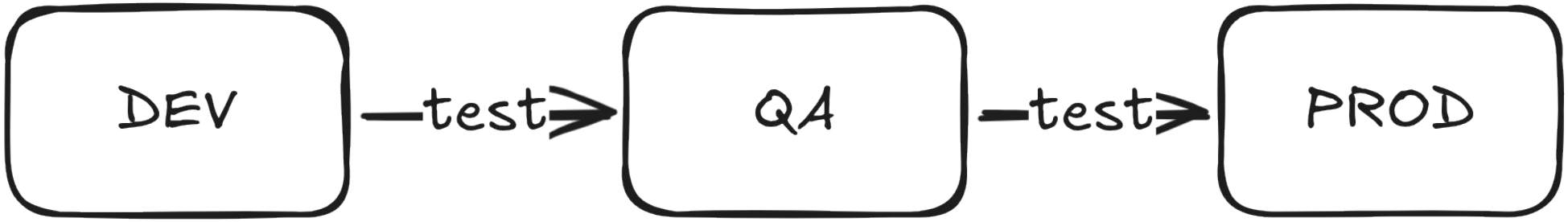
Static delays

Static delays

+ Low complexity
+ Basic safety
+ Good first step

- Emergencies
- Issue driven validation

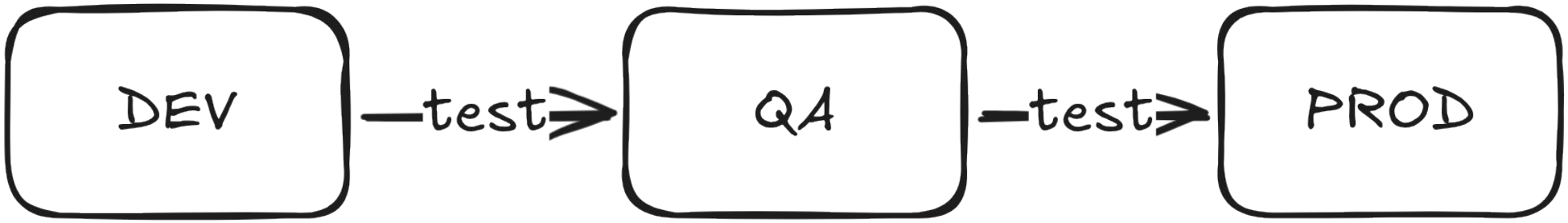
E2E-Gates



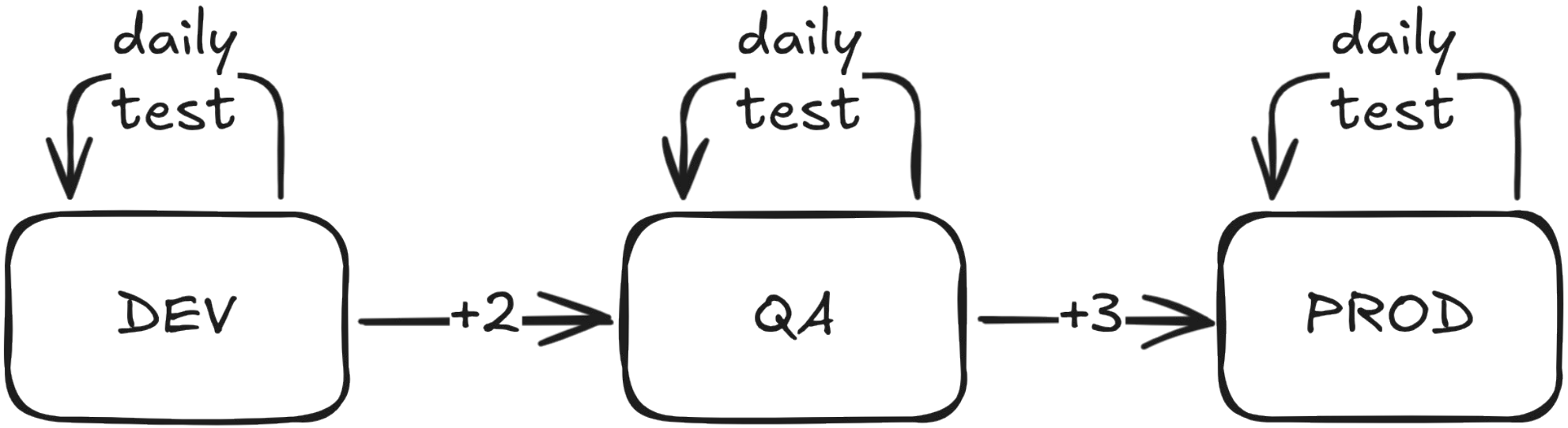
E2E-Gates

+ High confidence
+ Repeatable

- Complex to implement
- Needs maintainance
- 100%-CodeCov-Problem



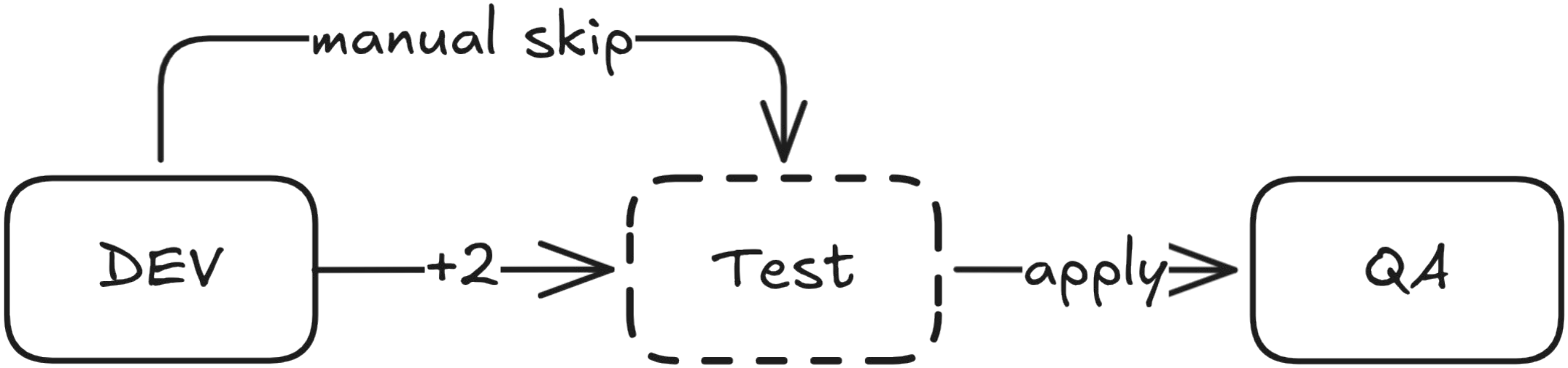
A healthy combination



Coping with reality

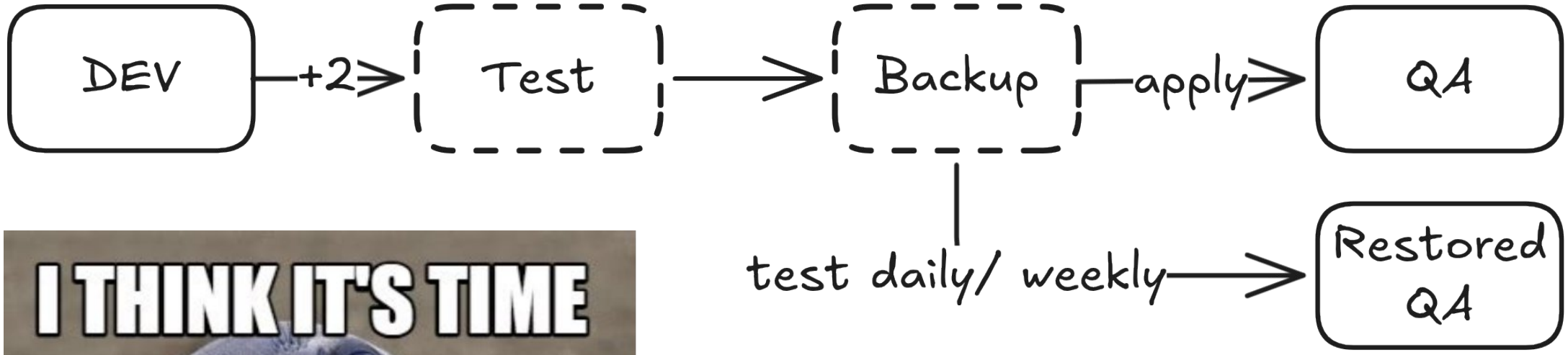
Something always breaks

Fast-Tracking



Backup/Restore

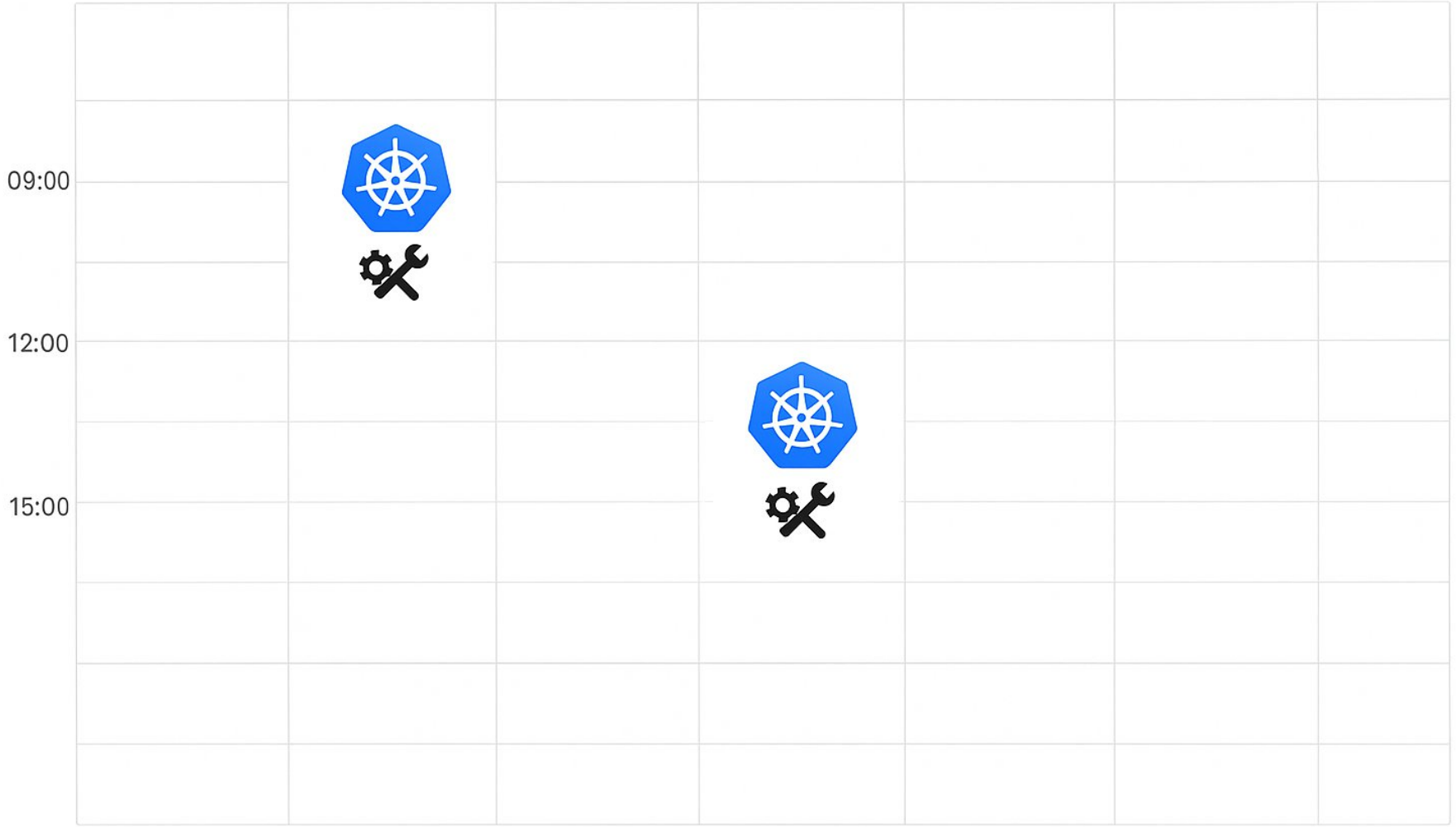
Backup/Restore



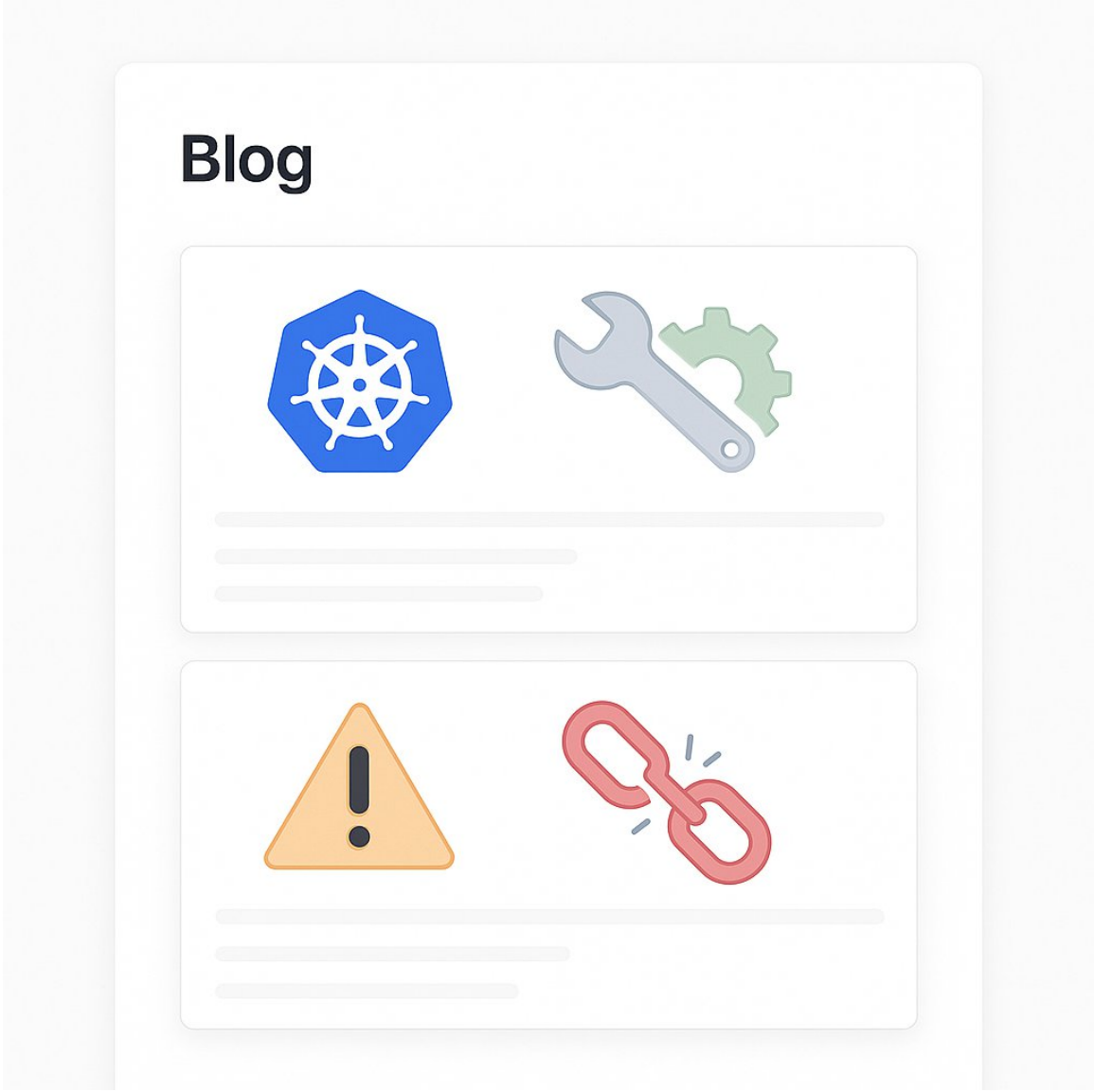
Additional

A quick sidequest in stakeholder communications

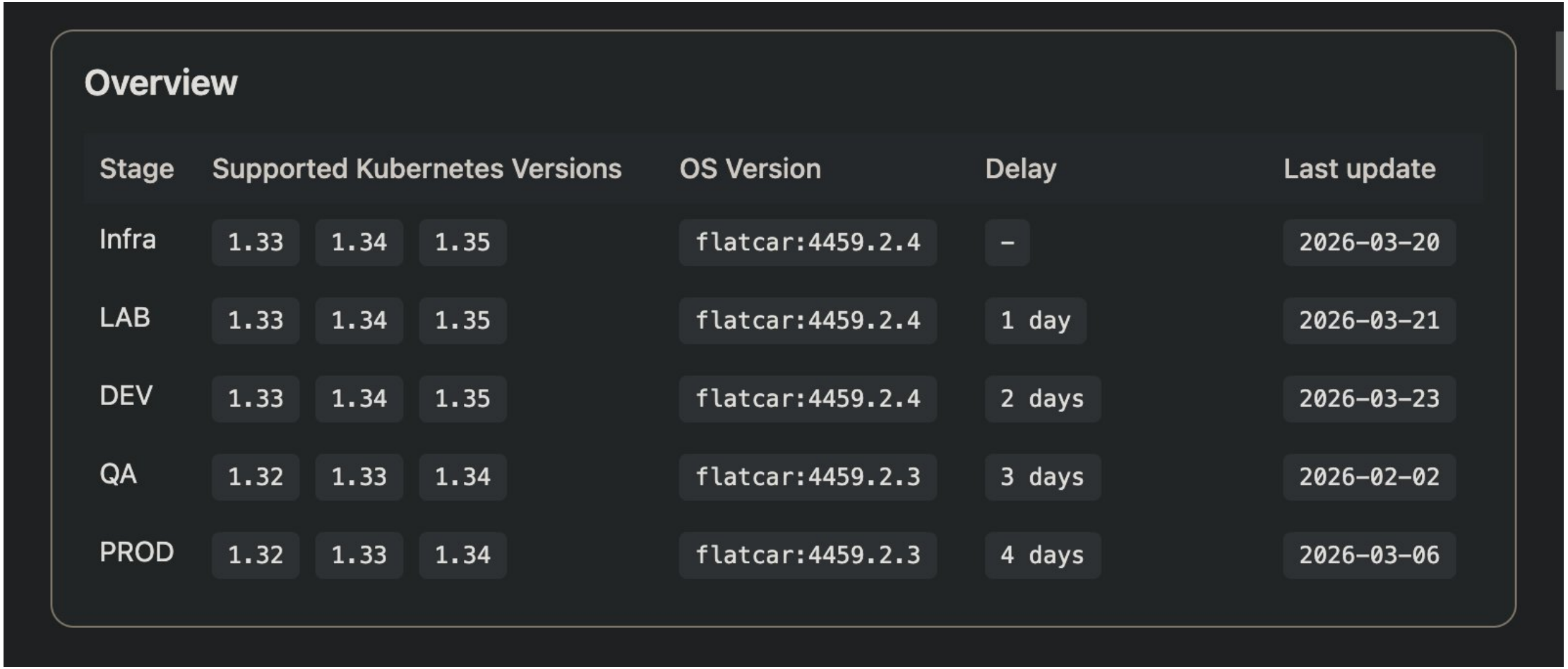
A likely baseline



Minor improvements



A recommendation



Overview

Stage	Supported Kubernetes Versions	OS Version	Delay	Last update
Infra	1.33 1.34 1.35	flatcar:4459.2.4	–	2026-03-20
LAB	1.33 1.34 1.35	flatcar:4459.2.4	1 day	2026-03-21
DEV	1.33 1.34 1.35	flatcar:4459.2.4	2 days	2026-03-23
QA	1.32 1.33 1.34	flatcar:4459.2.3	3 days	2026-02-02
PROD	1.32 1.33 1.34	flatcar:4459.2.3	4 days	2026-03-06

TL;DL

Stop yapping already

TL;DL

1. Find out what actually is a part of your platform

TL;DL

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2. Include all major dependencies in your staging

TL;DL

1. Find out what actually is a part of your platform
2. Include all major dependencies in your staging
3. Choose some kind of versioned state

TL;DL

1. Find out what actually is a part of your platform
2. Include all major dependencies in your staging
3. Choose some kind of versioned state
4. Start with delays and add automatic tests

TL;DL

1. Find out what actually is a part of your platform
2. Include all major dependencies in your staging
3. Choose some kind of versioned state
4. Start with delays and add automatic tests
5. Tell your users what is happening when to increase trust

Thank you!

Have a great
lunch break
Platform Engineering Day
KubeCon/CloudNativeCon

https://sched.co/2DY8r

Slides & More: https://github.com/nicolaiort/pedeu2026-promotion-failed

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